



Optimizing Fireplace Control Workflows for Immediate Implementation in Recalled Residential Gas Fireplaces

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Abstract

The purpose of this investigation is to redesign the Fireplace Control Circuit for the latest residential gas fireplace produced by the Acme Fireplace Company and recalled residential gas fireplaces. The scope of this research encompasses Fireplace Control Circuit design remediation in the United States.

The initial truth table will be derived from the Request For Proposal, and a corresponding logic expression and circuit design will be created. The National Instruments Multisim Circuit Design Software will be used to virtually design and simulate schematic/wiring diagrams of the Fireplace Control Circuit prototype to cement the prototype design prior to actual breadboarding. The final circuit will be implemented on the breadboard and every possible input will be tested multiple times to ensure the accuracy of the design.

The Fireplace Control Circuit implemented on the conventional protoboard with additional breadboards was 100% accurate throughout all trials and functioned with perfect precision. This design will increase consumer satisfaction, leading to decreasing residential gas fireplace accidents and increased profits for the Acme Fireplace Company upon commercial implementation. It is recommended that the client use the findings of this investigation to optimize the novel gas fireplace with the redesigned circuit.



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1.0 Introduction

1.1 Subject

Since their inception in 1895, gas fireplaces have been a staple of American homes. For modern fireplaces, the gas burner is controlled by an electronic circuit combined with thermal sensors. However, many of these fireplaces use circuits designed in the 20th century, and with advancements in circuit and sensor technology, the aforementioned circuits have become inefficient, resulting in the need for a more efficient replacement.

1.2 Purpose

The purpose of this investigation is to redesign the fireplace control circuit for the latest residential gas fireplace produced by the Acme Fireplace Company. The Acme Fireplace Company's new fireplace is designed to be more efficient, and thus, an additional purpose of this investigation is to design a more efficient circuit than the previous iteration.

1.3 Scope

The scope of this research encompasses several real-life applications. With the introduction of the Clean Air Act of 1956, there was an increased focus on automating gas fireplace systems; accompanying this focus are risks of burn hazards and possibilities of unnecessary Fireplace shutdowns or faulty sensor outputs, all of which tend to consumer dissatisfaction. Therefore, this study's results can be extrapolated towards Fireplace Control Circuit design remediation in the United States, especially in the wake of recent residential gas Fireplace recalls (e.g. Montebello ST- and DRT63ST-models by Innovative Hearth Products, LLC).

However, certain limitations exist within this research investigation: only the National Instruments (NI) Student Data Acquisition Device (abbr. NI myDAQ), NI Digilent myDigital Protoboard (abbr. NI Protoboard), and materials on hand that do not consider wear-and-tear conditions (e.g. wires, additional switches, etc.) for the Fireplace Control Circuit prototype are permitted; the prototype must be completed within four business days; and the prototype must be virtually simulated and approved before physical testing occurs.

1.4 Audience

This report is intended for the Directors of Panther Engineering so that the final Fireplace Control Circuit device may be assessed and implemented in the Acme Fireplace Company's residential gas fireplace base model, under the jurisdiction of Dauntless Industries, LLC. Alternatively, the findings of this study may be reviewed by the Circuits and Systems Society under the Institute of Electrical and Electronics Engineering (IEEE), to investigate greater applications of the Fireplace Control Circuit in other circuit control systems as a macro.

1.5 Development

The methods followed throughout this investigation will be presented first to provide the reader with a foundation of engineering standards followed. Any assumptions made that may impact the study will be discussed initially. The general methodology followed will be listed before the results of this study to guide the reader through the research investigation. The results will be accompanied with analyses of the preliminary and final



schematic diagrams created under the ‘Discussion’ section to justify the success of the design and will additionally allow reconstruction of this research study by the reader. Lastly, conclusions will be drawn from the Fireplace Control Circuit apparatus created and the Engineering Design Process, and references will be shown to enhance the reader’s comprehension on the subject matter.

2.0 Methods, Assumptions, and Procedures

2.1 Methods

All measurements are to be taken in accordance with the *Système International d’Unités* (SI) system of measurement, regulated by the *Bureau International des Poids et Mesures* (BIPM). Metric prefixes may be used for simplicity in computations.

The NI Multisim Circuit Design Software (CDS) will be used to virtually design and simulate schematic/wiring diagrams of the Fireplace Control Circuit prototype to cement the prototype design prior to actual breadboarding. The wiring diagrams will be used to allow for seamless breadboarding of the physical prototype of the Fireplace Control Circuit.

The problem was discovered through a Request for Proposal (RFP) provided by the Directors of Panther Engineering. All research conducted roughly mirrored the methodology in which civil/electrical engineering bids are solicited (through RFPs) and managed.

2.2 Assumptions

In this research investigation, functionality of the NI Multisim CDS in its adherence to Circuit Theory Laws and component properties is to be assumed.

Functionality of the NI myDAQ in its ability to transfer power from a computer to the NI Digilent myDigital Protoboard is to be assumed. This functionality is essential for the prototype to run and display and output values and hence will be assumed to facilitate progression of this research study.

Similarly, functionality of the NI Digilent myDigital Protoboard is to be assumed. In order to accurately develop the prototype for the Fireplace Control Circuit, the inputs from switches/buttons, outputs from lights, 5V power and Ground are required. The NI Protoboard’s functionality is essential to test and correct the Fireplace Control Circuit. As knowledge on the NI Protoboard’s internal mechanism workings that enable it to perform the aforementioned tasks is beyond the scope of this research investigation, it is to be assumed to allow the study to continue uninterrupted.

2.3 Procedures

1. Obtain the RFP from the Directors of Panther Engineering and read over the objectives of the project.
2. Create a Truth Table for the Fireplace Control Circuit based on the design specifications.
3. Create Karnaugh Maps for each output (i.e. the Emergency Control Valve and the Faulty Sensor Indicator) using the truth table and derive the simplified logic expressions for each output.



4. Create separate schematic designs for each output of the Fireplace Control Circuit based on the simplified logic expressions using an And-Or-Inverter (AOI) implementation per person (see Appendix C).
5. Convert the schematic design for the Emergency Control Valve from AOI to NAND implementation (see Appendix C).
6. Convert the schematic design for the Faulty Sensor Indicator from AOI to NOR implementation (see Appendix C).
7. Discuss the pros and cons of each design in the team to select favorable traits and ideas. Create a decision matrix, if applicable, to consolidate one final schematic design for the Emergency Control Valve and the Faulty Sensor Indicator.
8. Combine the final schematics into one detailed Final Schematic Design (see Appendix D) to model the circuit to be breadboarded.
9. Create a detailed wiring diagram to aid the Final Schematic Design (see Appendix D) and serve as a basis for breadboarding the prototype.
10. Gain approval from the Directors of Panther Engineering on the Final Schematic Designs to commence the building process.
11. Create the proposed circuit on the NI Digilent myDigital Protoboard by placing the required 14-pin Dual Inline Pin (DIP) Integrated Circuits (ICs) in the breadboard gap; proceed to carefully strip wires to meet the required length between breadboard holes according to the wiring diagram (See Appendices E and F). Troubleshoot any issues in the construction process by creating workarounds via changing inputs (e.g. change from switches to buttons or manually control power), changing circuit configuration (e.g. navigating around a breadboard row suspected to be defective), or the Test-Point Technique (using a probe to extract the logic expression after the truth table).
12. Provide a practical demonstration of the device created to the Directors of Panther Engineering to verify that all customer objectives and functionality are satisfied.
13. Draft a Technical Report presenting the findings of the study in accordance with the ANSI/NISO Z39.18-2005 (R2010) Technical Report standards.

3.0 Results and Discussion

3.1 Results

3.1.1 Overview

The Fireplace Control Circuit was redesigned per the aforementioned specifications outlined by the Acme Fireplace Company. The initial derived logic expression was deemed to be optimal after several reviews and a corresponding AOI circuit was designed by hand. After the circuit was deemed optimal, the circuit was then converted into a NAND/NOR implementation to optimize inventory and chip usage (see Appendix C). The sketched circuit was deemed to use an optimal number of 74LS00 and 74LS02 chips (see Appendix D). Two separate versions of the circuit were designed in the digital Circuit Design Software (CDS), after which the most efficient of the two designs was determined. The circuit was then designed and implemented on a conventional protoboard using additional breadboard attachments and tested for accuracy (see Appendix F).



The project was completed across the span of four business days, as described by the following subsections.

3.1.2 Day 01

The RFP from the Directors of Panther Engineering was obtained, and the objectives of the project were reviewed to gain a comprehensive understanding of the system requirements. Based on the specifications outlined in the RFP, a truth table for the Fireplace Control Circuit was developed to logically represent the system conditions. Subsequently, Karnaugh Maps for each output—the Emergency Control Valve and the Faulty Sensor Indicator—were created. The Karnaugh Maps were systematically analyzed to derive simplified logic expressions for both outputs in the circuit design.

3.1.3 Day 02

Using the simplified logic expressions derived on Day 1, separate schematic designs for each output of the Fireplace Control Circuit were constructed, following an AOI implementation. The AOI schematic for the Emergency Control Valve was reviewed and converted into a NAND gate implementation, while the Faulty Sensor Indicator schematic was converted into a NOR gate implementation. This conversion was conducted to optimize the circuit for chip availability and inventory usage. All resulting schematics were verified for logical accuracy and design integrity.

3.1.4 Day 03

A thorough evaluation of the NAND and NOR designs was conducted, discussing the advantages and disadvantages of each design. A decision matrix was developed to objectively assess each approach and consolidate favorable traits into one unified schematic design. The finalized schematics for both the Emergency Control Valve and the Faulty Sensor Indicator were combined into a single Final Schematic Design, ensuring compatibility and proper integration. A detailed wiring diagram was then created to provide a clear framework for the breadboarding process and minimize the potential for wiring errors.

3.1.5 Day 04

The Final Schematic Design and corresponding wiring diagram were reviewed and approved by the Directors of Panther Engineering. The prototype circuit was implemented on the NI Digilent myDigital Protoboard by placing the required 14-pin Dual Inline Pin (DIP) Integrated Circuits (ICs) on the conventional breadboard and wiring the inputs and outputs of the circuit. Wiring was conducted with precision, adhering to the detailed wiring diagram. Troubleshooting methods such as input adjustments, circuit reconfigurations, and the Test-Point Technique were employed to resolve any issues encountered during the construction process. Upon successful completion, a practical demonstration as well as multiple experimental trials of the device were performed to verify functionality and compliance with the client's objectives.

3.2 Discussion

The fireplace control circuit implemented on the conventional protoboard with the additional breadboard as described above was 100% accurate throughout all trials and



functioned with perfect precision (see Appendix A). The protoboard used was malfunctioning on its own, but due to a lack of sufficient resources, an additional protoboard was unable to be procured. To solve this problem, additional breadboards were wired to the protoboard as the only reliable components of the protoboard were the power supply and input buttons. The 74LS00 and 74LS02 chips were wired to the additional breadboards. With the additional breadboards, the circuit functioned optimally. The final design of the circuit can be justified by its perfect accuracy and precision, adeptly meeting the goals of the client.

4.0 Conclusion

The Final Design for the Fireplace Control Circuit surpassed all criteria while adhering to the limits set by the constraints. The configuration of the Final Design had yielded correct output values in accordance with the truth table for all configurations of inputs; the design was built upon 2 breadboards, equivalent in size/space to the NI Digilent myDigital Protoboard; and the constraint of time was also followed. The objectives set by the Directors of Panther Engineering were all accomplished in a timely manner: Circuit Theory Laws, NAND and NOR implementation over AOI logic, application of the Karnaugh Mapping Technique, and circuit (broadly, engineering) design properties were learned and applied to design a Fireplace Control Circuit satisfying all conditions of the original RFP/bid proposed by the Acme Fireplace Company. To that end, the results of the Final Design fulfill the problem and design statements.

If given a chance, the design solution would be created differently by trying to color-code inputs to their corresponding outputs to offer ease of use and troubleshooting from a consumer perspective as opposed to solely focusing on prototype functionality and efficiency; not only would this have emphasized user safety over product functionality, but it would increase consumer satisfaction, leading to decreasing residential gas fireplace accidents and increased profits for the Acme Fireplace Company (which would provide greater royalty profits for Dauntless Industries, LLC, and their local branch with Panther Engineering) upon commercial implementation, all of which would have increased prototype success in the overall duration of this investigation.

This research study presented itself as an opportunity to achieve greater proficiency with the NI Multisim CDS, and consolidated knowledge on Circuit Design techniques, including Karnaugh Mapping and converting AOI Logic to NAND and NOR implementations. Furthermore, an understanding of hardware inefficiencies and possibilities of error in standardized components (e.g. the NI Digilent myDigital Protoboard) was gained throughout the duration of this study.

Therefore, the Final Design was ultimately successful in surpassing all criteria/constraints and objectives placed by the Directors of Panther Engineering. An efficient and cost-effective Fireplace Control Circuit solution was created by this study, which can be extrapolated to real-world scenarios regarding safe and efficient automated residential gas Fireplace systems.

5.0 References

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Appendices

Appendix A: Truth Table

Table 1: Truth Table Input/Output Definitions

	0	1
A – Thermal Sensor	Sensor is off and does not indicate that a flame is present	Sensor is on and indicates that a flame is present
B – Thermal Sensor	Sensor is off and does not indicate that a flame is present	Sensor is on and indicates that a flame is present
C – Thermal Sensor	Sensor is off and does not indicate that a flame is present	Sensor is on and indicates that a flame is present
D – Thermal Sensor	Sensor is off and does not indicate that a flame is present	Sensor is on and indicates that a flame is present
E – Emergency Cutoff Valve	Valve is closed, no gas flows, and no flame is present	Valve is opened, gas flows, and a flame is present
F – Faulty Sensor Indicator	Sensors A, B, C, and D agree (report the same value: all 0 or all 1)	Sensors A, B, C, and D do not all agree (do not all report the same value)

Table 2: Truth Table and Empirical Data

A	B	C	D	E	F
0	0	0	0	0	0
0	0	0	1	0	1
0	0	1	0	0	1
0	0	1	1	0	1
0	1	0	0	0	1
0	1	0	1	0	1
0	1	1	0	0	1
0	1	1	1	1	1
1	0	0	0	0	1
1	0	0	1	0	1
1	0	1	0	0	1
1	0	1	1	1	1
1	1	0	0	0	1
1	1	0	1	1	1
1	1	1	0	1	1
1	1	1	1	1	0



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Appendix B: Karnaugh Maps and Logic Expressions

Table 3: Karnaugh Map for Output E

	$\bar{C}\bar{D}$	$\bar{C}D$	$C\bar{D}$	CD
$\bar{A}\bar{B}$	0	0	0	0
$\bar{A}B$	0	0	1	0
AB	0	1	1	1
$A\bar{B}$	0	0	1	0

$$E = BCD + ACD + ABD + ABC$$

Table 4: Karnaugh Map for Output F

	$\bar{C}\bar{D}$	$\bar{C}D$	$C\bar{D}$	CD
$\bar{A}\bar{B}$	0	1	1	1
$\bar{A}B$	1	1	1	1
AB	1	1	0	1
$A\bar{B}$	1	1	1	1

$$F = \bar{A}D + B\bar{C} + A\bar{B} + C\bar{D}$$



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Appendix C: Preliminary Schematics

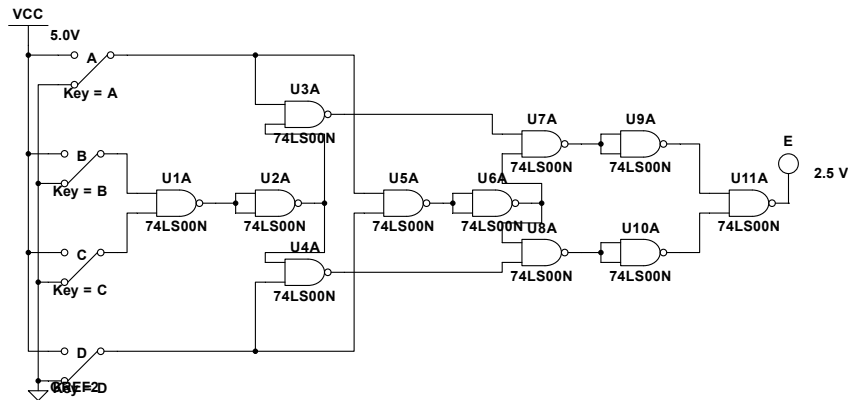


Figure 1: V. Sundar and H. Chandran's Preliminary Schematic for Output E

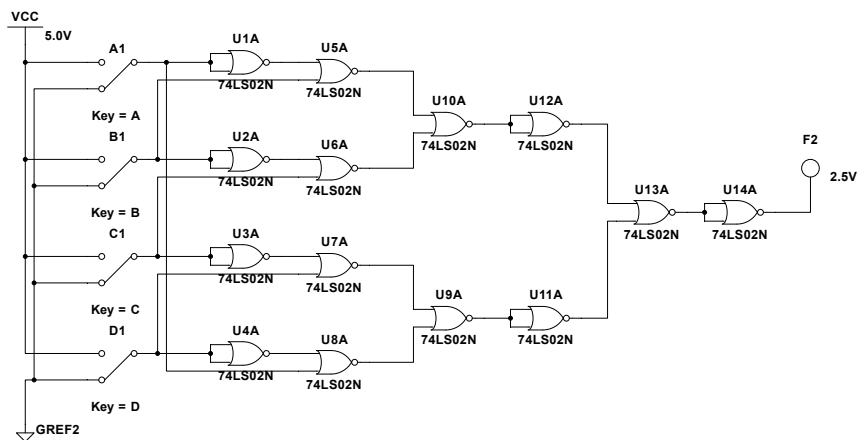


Figure 2: V. Sundar's Preliminary Schematic for Output F

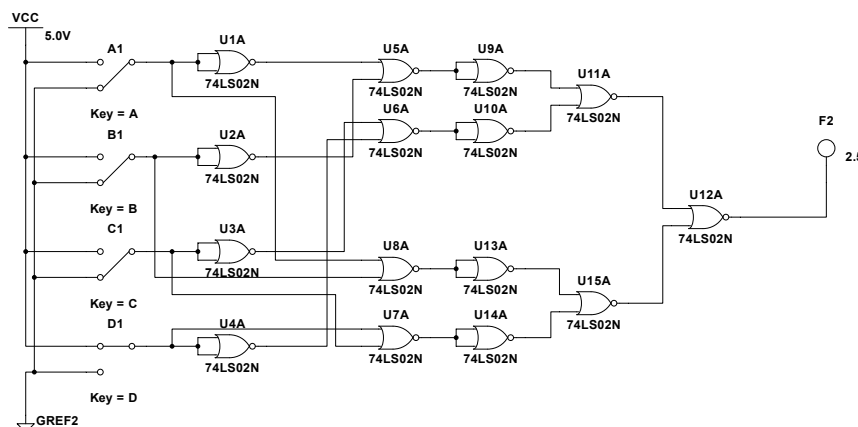


Figure 3: H. Chandran's Preliminary Schematic for Output F



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Appendix D: Final Schematics

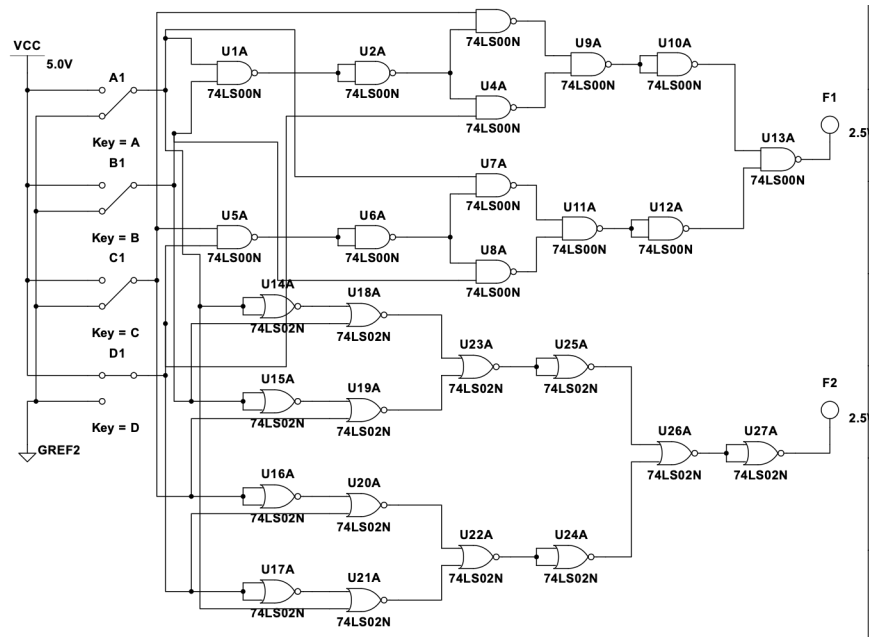


Figure 4: Final Schematic Design

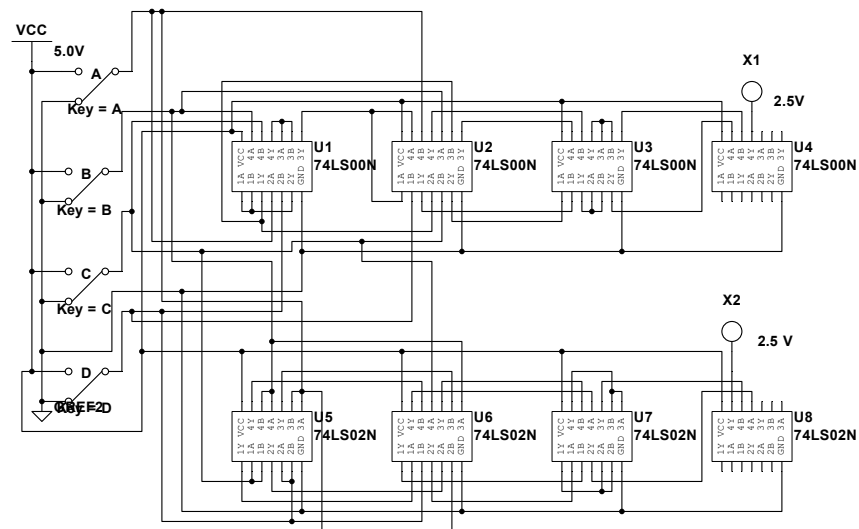


Figure 5: Final Schematic Wiring Diagram



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Appendix E: Parts List

4 – 74LS00 14-Pin DIP ICs

4 – 74LS02 14-Pin DIP ICs

88 – Jumper Wires (Custom length)

1 – National Instruments 192256A-01 Type USB A/B 2.0 Cable (1 meter)

1 – National Instruments Digilent myDigital Protoboard

1 – National Instruments Student Data Acquisition Module

2 – Zhong Yi Electronic MB-102 Solderless Breadboard



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Appendix F: Final Circuit Photograph

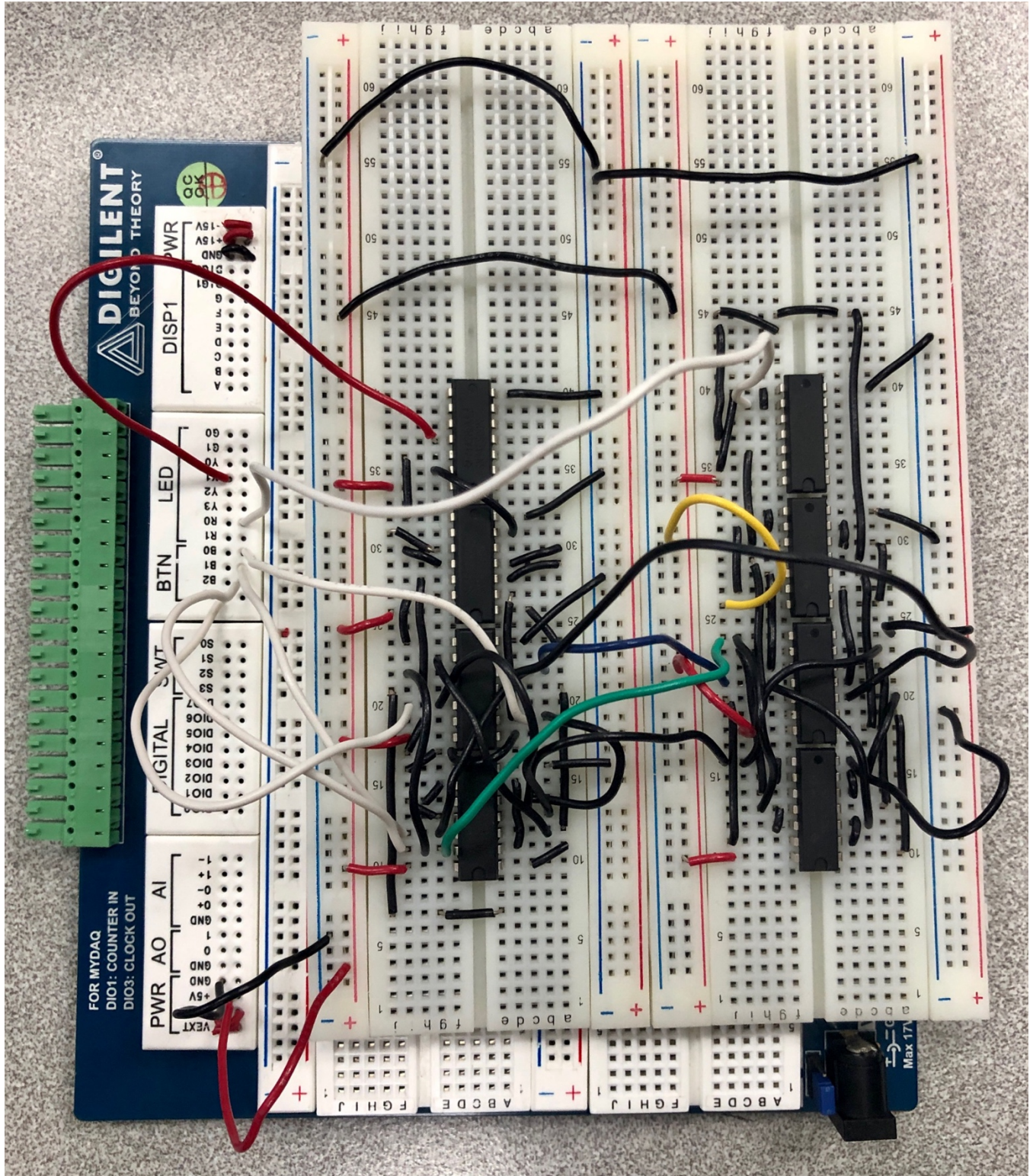


Figure 6: Final Circuit Photograph



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List of Symbols, Abbreviations, and Acronyms

Symbols

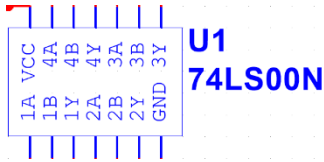


Figure 7: 74LS00 14-Pin DIP IC Schematic Symbol

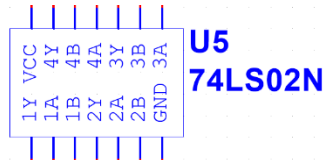


Figure 8: 74LS02 14-Pin DIP IC Schematic Symbol



Figure 9: Ground (GND) Pin-Out Schematic Symbol

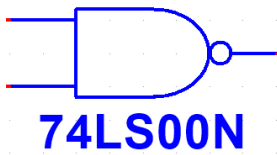


Figure 10: 2-Input 74LS00 NAND Gate Schematic Symbol



Figure 11: 2-Input 74LS02 NOR Gate Schematic Symbol

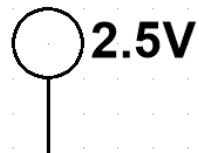


Figure 12: Digital Probe Schematic Symbol



Figure 13: Single Pole Double Throw (SPDT) Switch Schematic Symbol

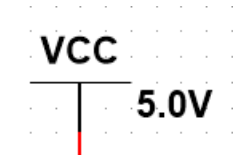


Figure 14: Voltage Common Collector (VCC) Schematic Symbol



Abbreviations

NAND – Not-AND

NI myDAQ – National Instruments Student Data Acquisition Module

NOR – Not-OR

NI Protoboard – National Instruments Digilent myDigital Protoboard

Acronyms

ANSI – American National Standards Institute

AOI – AND-OR-Invert Logic

BIPM – International Bureau of Weights and Measures

CDS – Circuit Design Software

DAQ – Data Acquisition Module

DIP – Dual Inline Pin

DRT – Direct Vent

GND – Ground

IC – Integrated Circuit

IEEE – Institute of Electrical and Electronics Engineers

LLC – Limited Legal Corporation

NI – National Instruments

NISO – National Information Standards Organization

RFP – Request For Proposal

SI – International System of Units

SPDT – Single Pole Double Throw

ST – See-Through

VCC – Voltage Common Collector